Nexio CRM: TZ va joriy loyiha solishtirmasi

Manba: Nexio_CRM_Full_Plan_with_8_Screens.pdf (v2.0, 12.10.2026). Loyiha: nexio_crm_v2 (main, 43a9031). Tekshiruv sanasi: 09.11.2026. Baholash kod (React src/, Laravel backend/) bo'yicha statik tahlil asosida; ishga tushirib sinalmagan.

Jami talablar: 54 | **Bor: 19** | **Qisman: 26** | **Yo'q: 9**

1. 8 ta asosiy ekran (PDF 2-9 betlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
01 Dashboard	**Bor**	Dashboard.tsx + GET /dashboard: KPI, savdo grafigi (Recharts), voronka, vazifalar, AI faollik, so'nggi lid/buyurtmalar; qo'shimcha MyDay sahifasi	KPI kartalaridan filtrli o'tish hali yo'q (davr filtri URL'da saqlanadi - YANGI)
02 Login / Welcome	**Bor**	Login, Register, parolni tiklash, email tasdiqlash, taklif (invite), workspace almashtirish (switch-tenant); YANGI: MFA/TOTP (RFC 6238 testlangan; sozlamalar > Xavfsizlik; resurs kodlari, qayta ishlatishdan va brute-force'dan himoya), /choose-workspace (bir nechta kompaniya bo'lsa login keyin), /onboarding (haqiqiy ma'lumotga asoslangan 4 qadamli checklist, ro'yxatdan keyin)	Alohida /mfa marshruti yo'q (kod kiritish login sahifasida ikkinchi bosqich sifatida)
03 Leads & Pipeline	**Qisman**	Lidlar ro'yxati, yaratish, konvertatsiya, diskvalifikatsiya; Deals kanban (drag&drop), /deals/:id/move, CSV eksport; YANGI: CSV import wizard (ko'rib chiqish + validatsiya + formula-inyeksiya himoyasi), /app/leads/:id va /app/deals/:id detail sahifalari	dnd-kit o'rniga native DnD (kutubxona o'rnatilmagan)
04 Clients / Companies	**Qisman**	Companies, CompanyDetail (tablar), Contacts, faoliyat (activities); YANGI: /app/contacts/:id detail sahifasi	Custom fields yo'q; contact_company (ko'p-kompaniya) jadvali yo'q; notes/files tablari yo'q
05 Projects & Orders	**Qisman**	Projects, ProjectDetail, milestones, Tasks, Orders, buyurtma tasdiqlash, buyurtmadan loyiha yaratish; YANGI: Gantt (loyihalar vaqt shkalasi), /app/calendar (vazifa/loyiha/bitim sanalari), /app/orders/:id detail sahifasi	Fayllar/attachments, time entries, task dependencies yo'q
06 AI Control Center	**Qisman**	AI agentlar (CRUD, status), kampaniya start/stop, qo'ng'iroqlar ro'yxati/tafsilot, approvals, suppression list, AI analitika, Groq LLM, telephony plagini	Real ovozli streaming (ASR/TTS) yo'q - qo'ng'iroqlar sandbox/simulate; knowledge base faqat JSON maydon; alohida Policies, Usage, Campaign detail sahifalari yo'q
07 Billing & Subscriptions	**Qisman**	Tariflar, tarif almashtirish, bekor/qayta tiklash, invoice to'lash, Click webhook (imzo tekshiruvi, dublikatdan himoya), avto-uzaytirish komandasi	Payme yo'q; to'lov usuli/tranzaksiya tarixi alohida sahifasi yo'q; add-on modullar savdosi qisman (marketplace orqali)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
08 Platform Super Admin	Bor	Alohida PlatformShell: overview, tenants, plans, subscriptions, usage, Nexio buyurtmalar, support, AI-ops, security, settings, marketplace, audit API	/platform/health sahifasi va health-check monitoringi yo'q

2. Qo'shimcha marshrutlar (PDF 4-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Auth marshrutlari	Bor	/login, /register, /reset-password, /verify-email, /invite/:token, YANGI: /choose-workspace, /onboarding	Alohida /mfa marshruti (funksional jihatdan login ichida)
CRM detail sahifalari	Bor	/app/companies/:id, YANGI: /app/leads/:id, /app/contacts/:id, /app/deals/:id	-
Operatsiyalar	Qisman	YANGI: calendar; orders, YANGI: orders/:id, projects/:id, tasks, communications, helpdesk (support), documents	/app/files
Moliya	Qisman	/app/finance (umumiy), /app/billing, /app/catalog (products)	Alohida /app/invoices, /app/payments, /app/expenses, /app/usage
Hisobot va avtomatlashtirish	Qisman	/app/analytics, /app/automations (dry-run bilan), /app/integrations	/app/reports, /app/api-keys, /app/webhooks
AI marshrutlari	Qisman	/app/ai-agents, /ai-agents/:id, /ai-calls, /ai-analytics	/ai/campaigns/:id, /ai/knowledge, /ai/usage, /ai/policies
Jamoa va sozlamalar	Qisman	/app/team (rollar/ruxsatlar), /app/settings (logo, brend rangi, til, vaqt zonasi)	Alohida /app/roles va /app/settings/security
Nexio xizmatlari / portal	Bor	/app/nexio-services, mijoz portali /portal	-

3. Dizayn-tizim (PDF 3-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rang tokenlari	Qisman	#2563EB / #1D4ED8 va boshqa ranglar CSS o'zgaruvchilarida (--color-*)	--nx-* nomlash standarti yo'q; qo'shimcha tungi tema mavjud (TZ'da faqat light)
Inter shrifti	Qisman	font-family: Inter, system-ui ...	Self-hosted WOFF2 Inter ulanmagan (faqat tizimdagi bo'lsa ishlaydi)
Lucide ikonkalar, Recharts	Bor	lucide-react, recharts	-
Holatlar: skeleton, empty, error, toast	Bor	UI komponentlar: core, table, toast, overlay, form	prefers-reduced-motion qo'llab-quvvatlashi topilmadi
RU / UZ / EN interfeys	Bor	O'z i18n provayderi (en/ru/uz), standart: uz	i18next ishlatilmagan (TZ talabi); barcha sahifalar to'liq tarjima qilinganini tekshirish kerak

4. Frontend steki (PDF 5-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
React + TS + Vite + Router	**Bor**	React 18, TypeScript, Vite 6, react-router 6, lazy loading	-
TanStack Query, Tailwind, Zod	**Bor**	@tanstack/react-query, Tailwind 4, zod	-
Zustand, shadcn/Radix, RHF, TanStack Table, dnd-kit	**Yo'q**	O'zimizning UI komponentlar va native DnD	Ushbu kutubxonalar o'rnatilmagan
Testlar (Vitest, RTL, Playwright)	**Qisman**	Vitest: 3 ta unit test (api, format, timezones)	React Testing Library va Playwright e2e yo'q
Kod tuzilmasi (apps/web, entities, features, packages/ui)	**Yo'q**	src/pages, src/components, src/lib	Monorepo, alohida apps/admin, packages/ui, OpenAPI client yo'q

5. Backend - Laravel (PDF 6-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
PHP 8.3 + Laravel + PostgreSQL	**Bor**	Laravel 13, PHP ^8.3, DB_CONNECTION=pgsql (SqlDialect orqali)	-
Redis, Horizon, S3	**Yo'q**	Queue/cache = database	Redis, Horizon, S3 obyekt saqlash sozlanmagan
Sanctum / OAuth2	**Yo'q**	O'z sessiya mexanizmi (sessions jadvali, ApiContext middleware)	Sanctum, OAuth2 third-party yo'q
Modul tuzilmasi (Policy, FormRequest, Event, Job)	**Qisman**	Domen bo'yicha controllerlar + Rbac, H, Ctx yordamchilari; Plugin tizimi (EventBus, Lifecycle)	Policies, FormRequests, Repository/Event/Job qatlamlari yo'q; controllerlar katta
Identity / Tenancy	**Bor**	Users, tenants, memberships, invitations, switch-tenant, 14 tenant roli, MFA	-
CRM / Sales	**Qisman**	Organizations, contacts, leads, pipelines, deals, activities, products, orders	Quotes, price_lists, tags, custom fields, notes yo'q
Projects	**Qisman**	Projects, milestones, tasks	Project members, comments, attachments, time entries yo'q
Finance / Subscription	**Qisman**	Invoices, payments, plans, subscriptions, usage_records, billing_events, prorating, past_due/grace	Expenses, credit notes, refund, currency_rates jadvali yo'q (CBU kurs servisi bor)
Integration	**Qisman**	Integrations toggle, webhook_events, Telegram bot, plugin marketplace	API tokenlar (scoped), outbound webhooks (HMAC, retry, DLQ) yo'q
Audit	**Bor**	audit_logs, /settings/audit, plugin audit	-
OpenAPI spetsifikatsiya	**Yo'q**	-	OpenAPI 3.1 schema, Postman/Bruno kolleksiya yo'q

6. Multi-tenancy, rollar va xavfsizlik (PDF 9, 14-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
tenant_id izolyatsiyasi	**Bor**	Barcha so'rovlarda tenant_id filtri	PostgreSQL RLS yo'q
Rollar va ruxsatlar matritsasi	**Bor**	Rbac.php: tenant_owner, admin, sales_manager, finance_manager va b.	Scope (own/team/all) darajasini tekshirish kerak
Tenant isolation testlari	**Qisman**	PluginTenantIsolationTest + 9 ta boshqa backend test	CRM uchun to'liq A/B tenant izolyatsiya testlari yo'q
MFA, rate limit, CSP/HSTS	**Qisman**	YANGI: MFA (TOTP) + MFA urinishlariga limit (5 ta / 5 daqiqa); resend-verification throttle	Umumiy login rate limit, CSP/HSTS sozlamalari topilmadi

7. Nexio AI Agents (PDF 11-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rozilik va DNC	**Bor**	consent_calls, suppression_list, do_not_contact, compliance gate	-
Qo'ng'iroq state machine	**Qisman**	queued, compliance_checked, dialing, connected, ai_disclosed, conversation ...	Provider event id bilan idempotent callbacklar real provayderda tekshirilmagan
Voice Sales Agent (real qo'ng'iroq)	**Yo'q**	Telephony plagini (HTTP provider + sandbox), simulate call	Real SIP/telefoniya + ASR/LLM/TTS streaming gateway yo'q
Buyurtmani AI tasdiqlashi + approval	**Qisman**	ai_approvals, finalize call, ai_tool_calls	Narx/chegirma limitlari to'liq tekshiruvini tasdiqlash kerak
Xarajat limitlari	**Qisman**	Agentda budget_limit	Kampaniya/kun/oy bo'yicha hard stop tekshirilmagan
Analytics/Support/Ops agentlari	**Qisman**	AI analitika sahifasi, AI assistant plagini (Groq)	Alohida Support va Ops agentlari yo'q

8. Integratsiyalar, DevOps va hujjatlar (PDF 13, 14, 18-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Telegram	**Bor**	Bot, webhook, polling, foydalanuvchini bog'lash	-
To'lov provayderlari	**Qisman**	Click (imzo bilan), sandbox	Payme yo'q
Web-forma SDK / widget	**Yo'q**	-	Saytdan lid olish widgeti yo'q
Docker Compose, CI/CD	**Yo'q**	deploy.sh skripti	docker-compose, Dockerfile, GitHub Actions yo'q
Monitoring (Sentry, health, alerts), backup	**Yo'q**	-	Sentry, alerting, backup/restore runbook yo'q
Hujjatlar	**Qisman**	PLUGIN_ARCHITECTURE.md, NEXIO_INVENTORY*.md, .env.example, seederlar	ARCHITECTURE, SECURITY, TENANCY, OPERATIONS, AI_POLICIES .md, RU/UZ/EN qo'llanmalar yo'q

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
TZ'dan tashqari qo'shimchalar	**Bor**	Plugin marketplace + developer portal, Inventory (ombor), Delivery (Uzum, Yandex), Helpdesk, Documents (taklif generatsiyasi), Automations	-

Asosiy xulosa: MVP uchun eng muhim yetishmayotganlar

1. (Bajarildi) MFA, /onboarding, /choose-workspace.
2. To'liq tenant izolyatsiya testlari (CRM uchun A/B tenant) - TZ bo'yicha release gate.
3. Custom fields, kontakt-kompaniya (ko'p-ko'p) bog'lanishi, notes/files tablari.
4. Fayllar (attachments), time entries, task dependencies, quotes (tijorat taklifi) moduli.
5. Scoped API tokenlar, outbound webhooks (HMAC), OpenAPI spetsifikatsiya.
6. Docker Compose, CI/CD, monitoring (Sentry/health), backup; Redis/queue worker.
7. AI: real telefoniya + ovozli gateway (hozircha sandbox/simulate), Payme integratsiyasi.